21 ELECTRIC MOTORS AND ELECTROMAGNETIC INDUCTION

Trains in many cities use electric motors to transport millions of people to and from work each day. In the first part of this chapter, we are going to look at how an electric motor works. An electric motor creates motion from an electric current. Generators and alternators, on the other hand, transfer movement or kinetic energy into electrical energy by a process called electromagnetic induction. The second part of this chapter is going to look at this process.



▲ Figure 21.1 Electric train in Bangkok (the Skytrain)

LEARNING OBJECTIVES

PHYSICS ONLY

- Know that there is a force on a charged particle when it moves in a magnetic field as long as its motion is not parallel to the field
- Understand why a force is exerted on a current-carrying wire in a magnetic field, and how this effect is applied in simple d.c. electric motors and loudspeakers
- Use the left-hand rule to predict the direction of the resulting force when a wire carries a current perpendicular to a magnetic field
- Describe how the force on a current-carrying conductor in a magnetic field changes with the magnitude and direction of the field and current
- Know that a voltage is induced in a conductor or a coil when it moves through a magnetic field or when a magnetic field changes through it and describe the factors that affect the size of the induced voltage
- Describe the generation of electricity by the rotation of a magnet within a coil of wire and of a coil of wire within a magnetic field, and describe the factors that affect the size of the induced voltage

PHYSICS ONLY

- Describe the structure of a transformer, and understand that a transformer changes the size of an alternating voltage by having different numbers of turns on the input and output sides
- Explain the use of step-up and step-down transformers in the large-scale generation and transmission of electrical energy
- Know and use the relationship between input (primary) and output (secondary) voltages and the turns ratio for a transformer:

$$\frac{\text{input (primary) voltage}}{\text{output (secondary) voltage}} = \frac{\text{primary turns}}{\text{secondary turns}}$$

- Know and use the relationship:

input power = output power

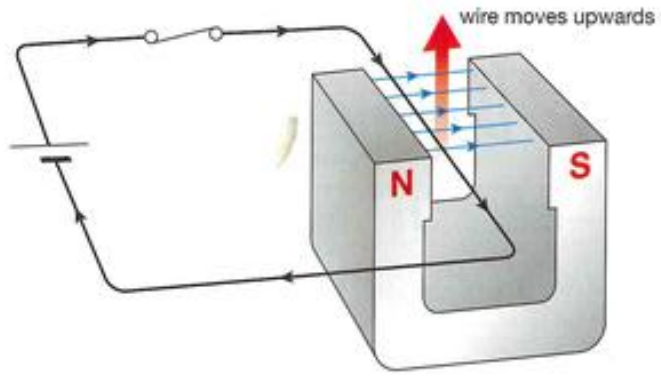
$$V_p I_p = V_s I_s$$

for 100% efficiency

MOVEMENT FROM ELECTRICITY

When a charged particle moves through a magnetic field it experiences a force, as long as its motion is not parallel to the field. We can demonstrate this effect by passing electrons (that is, an electric current) along a wire that is placed in a magnetic field as shown in Figure 21.2. When the switch is closed and electrons flow, the wire will try to move upwards.

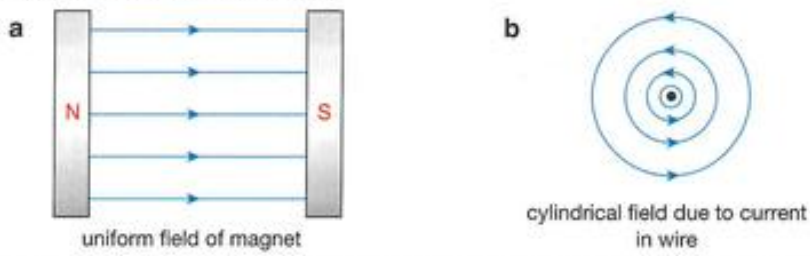
OVERLAPPING MAGNETIC FIELDS



▲ Figure 21.2 The wire moves where there is a current in it.

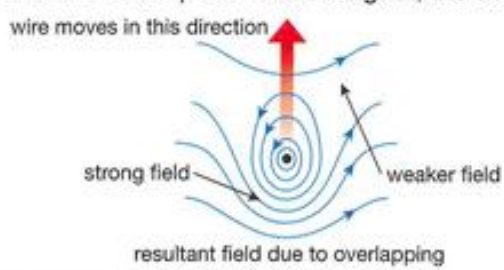
We can explain this motion by looking at what happens when the switch is closed and the two magnetic fields overlap.

The two diagrams in Figure 21.3 show the shapes and directions of the magnetic fields **a** across the poles of the magnet, and **b** around the current-carrying wire before they overlap.



▲ Figure 21.3 a Magnet's uniform magnetic field b Magnetic field around a current carrying wire

If the wire is placed between the poles of the magnet, the two fields overlap.



▲ Figure 21.4 The new magnetic field created by overlapping

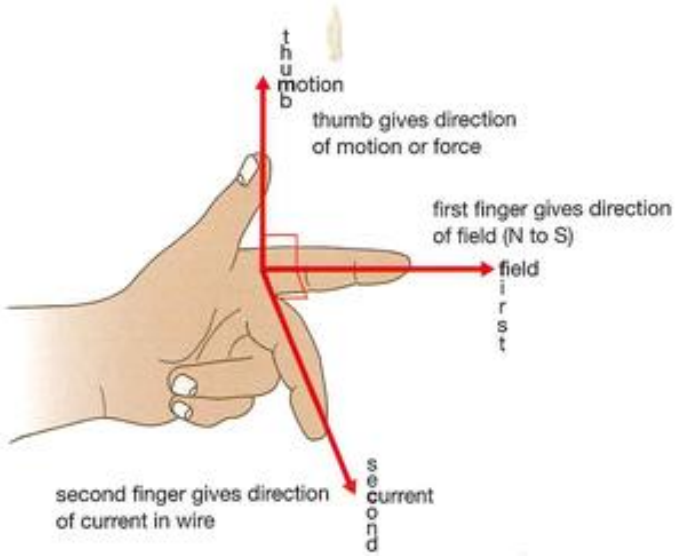
In certain places, for example, below the wire, the fields are in the same direction and so reinforce each other. A strong magnetic field is produced here. In other places, for example, above the wire, the fields are in opposite directions. A weaker field is produced here. Because the fields are of different strengths the wire 'feels' a force, pushing it from the stronger part of the field to the weaker part – that is, in this case, upwards. The overlapping of the two magnetic fields has produced motion. This is called the motor effect.

A stronger force will be produced if the magnetic field is stronger or if the current is increased.

KEY POINT

We can predict the direction of the force or movement of the wire using Fleming's left-hand rule.

If we change the direction of the current or the magnet's field, a different overlapping pattern is created and we will see the wire move in the opposite direction.

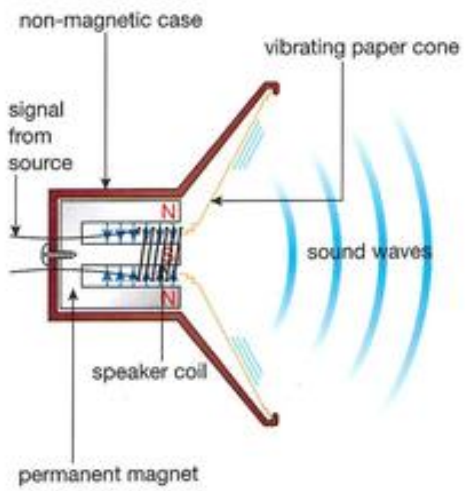


▲ Figure 21.5 Fleming's left-hand rule helps you to work out the direction of the force.

THE MOVING-COIL LOUDSPEAKER

The moving-coil loudspeaker uses the motor effect to transfer electrical energy to sound energy.

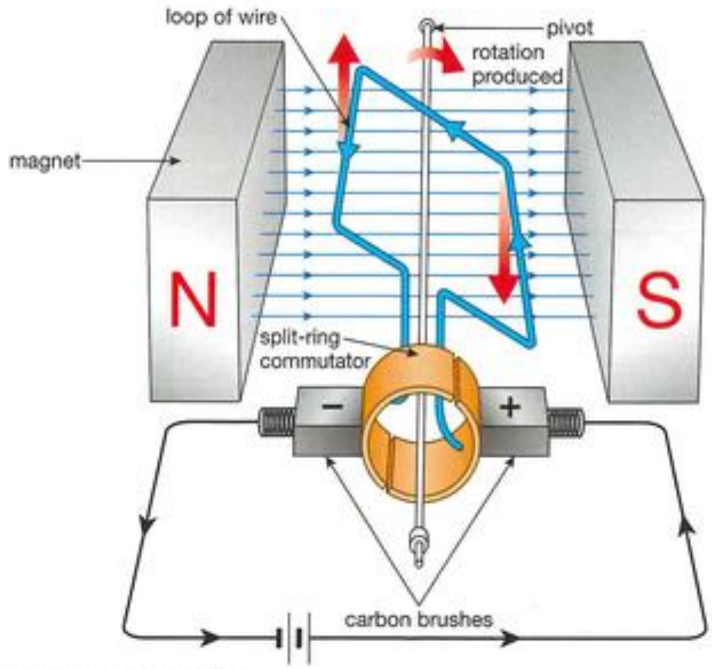
- Electric currents from a source, such as a radio, pass through the coils of a speaker.
- These currents, which represent sounds, are always changing in size and direction, like vibrating sound waves.
- The fields of the coil and the permanent magnet are therefore creating magnetic field patterns which are also always changing in strength and direction.
- These fields in turn apply rapidly changing forces to the wires of the coil, which cause the speaker cone to vibrate.
- These vibrations create the sound waves we hear.



▲ Figure 21.6 A loudspeaker transfers electrical energy to sound energy.

THE ELECTRIC MOTOR

The electric motor is one of the most important uses of the motor effect. Figure 21.7 shows the most important features of a simple d.c. electric motor.



▲ Figure 21.7 A simple electric motor

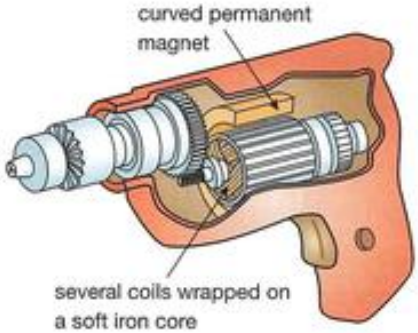
When there is current in the loop of wire, one side of it will experience a force pushing it upwards. The other side will feel a force pushing it downwards, so the loop will begin to rotate (turn).

As the loop reaches the vertical position, its momentum takes it past the vertical. If the rotation is to continue the forces on the wires must now be reversed so that the wire at the top is now pushed down and the bottom one is pushed up.

This can be done easily by using a split ring to connect the loop of wire to the electrical supply. Now each time the loop of wire passes the vertical position, the connections change, the direction of the current changes, and the forces on the different sides of the loop change direction. The loop will rotate continuously.

To increase the rate at which the motor turns we can:

1 increase the number of turns or loops of wire, making a coil
2 increase the strength of the magnetic field
3 increase the current in the loop of wire.



▲ Figure 21.8 A real electric motor

EXTENSION WORK

Although you will not be asked this in your exam it is interesting to know how commercial motors like the one used in this electric drill differ from the simple motor described in Figure 21.7.

- The permanent magnets are replaced with curved electromagnets capable of producing very strong magnetic fields.
- The single loop of wire is replaced with several coils of wire wrapped on the same axis. This makes the motor more powerful and allows it to run more smoothly.
- The coils are wrapped on a soft iron core that has been covered by a thin layer of plastic. This makes the motor more efficient and more powerful.

ELECTROMAGNETIC INDUCTION

Motors use electricity to produce movement. Generators and alternators are machines that use motion to produce electricity. They use a process called electromagnetic induction.

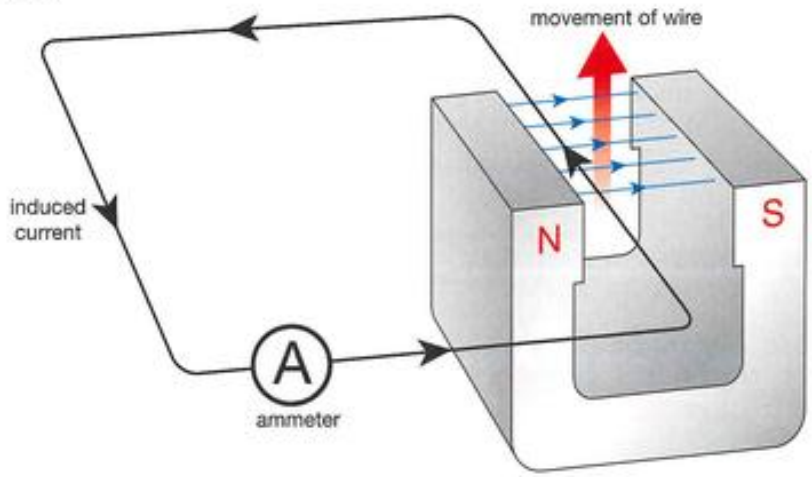


▲ Figure 21.9 Transportable generators are used to produce electricity for the lights and machinery used on roadworks.

The workers shown in Figure 21.9 need electricity for their machines and their lights. Instead of connecting into the mains supply, as we do at home, the workers have their own generator, which produces the electrical energy they need. In fact, the mains supply itself is produced by large generators in power stations. In this next section, you will discover how a generator produces electricity.

DEMONSTRATING ELECTROMAGNETIC INDUCTION

If we move a wire across a magnetic field at right angles, as shown in Figure 21.10, a voltage is induced or generated in the wire. If the wire is part of a complete circuit, there is a current. This event is called electromagnetic induction.

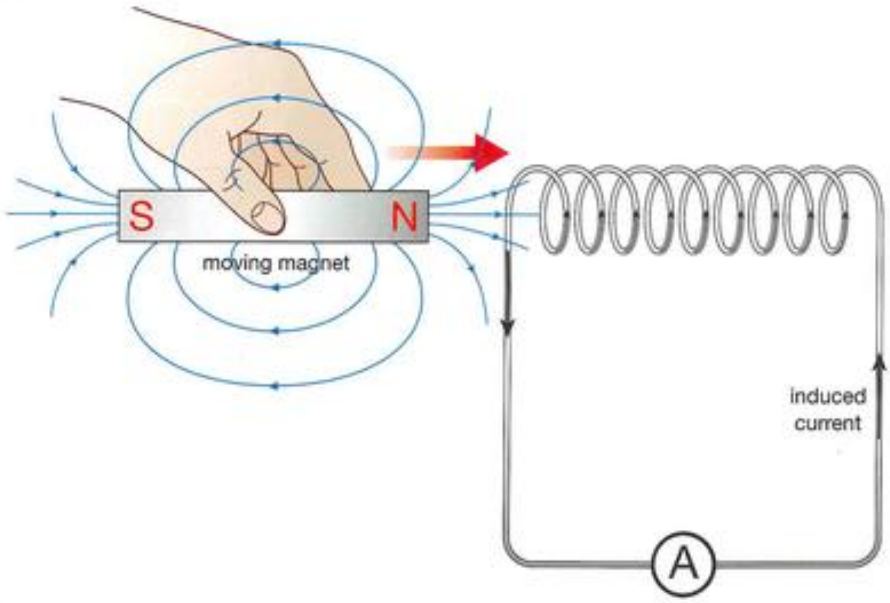


▲ Figure 21.10 When a wire moves across a magnetic field a voltage is generated in the wire.

The size of the induced voltage (and current) can be increased by:

1 moving the wire more quickly
2 using a stronger magnet so that there are more field lines 'cut'
3 wrapping the wire into a coil so that more pieces of wire move through the magnetic field.

We can also generate a voltage and current by pushing a magnet into a coil. In this situation and the previous one (Figure 21.10) we can see that it is the cutting action between the wires and the field lines that generates the voltage. If there is no cutting (that is, the wires and magnets are stationary) no voltage is generated.



▲ Figure 21.11 A magnet moving in a coil will generate electricity.

This experiment also shows us that the size of the induced voltage (and current) can be increased by:

1 moving the magnet more quickly

2 using a stronger magnet

3 using a coil with more turns.

We can summarise all the discoveries from these experiments by saying:

- a voltage is induced when a conductor cuts through magnetic field lines
- a voltage is induced when magnetic field lines cut through a conductor
- the faster the lines are cut the larger the induced voltage.

EXTENSION WORK

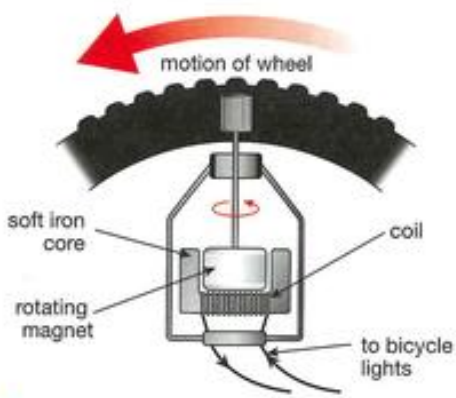
It is interesting to note, but not necessary for your exam, that Michael Faraday was the first person to observe how the size of an induced voltage depends upon the rate at which the magnetic lines of flux (field lines) are being cut. He summarised his observations in Faraday's Law of Electromagnetic Induction. This states that:

The size of the induced voltage across the ends of a wire (coil) is directly proportional to the rate at which the magnetic lines of flux are being cut.

GENERATORS

Figure 21.12 shows a small generator or dynamo used to generate electricity for a bicycle light.

As the cyclist pedals, the wheel rotates and makes a small magnet within the dynamo turn around. As this magnet turns, its magnetic field turns too. The field lines cut through the coil inducing a current in it. This current can be used to work the cyclist's lights.



▲ Figure 21.12 A dynamo is a simple generator.

Figure 21.13 shows a much larger generator used in power stations to generate the mains electricity we use in our homes. In these generators the magnets are stationary and the coils rotate.



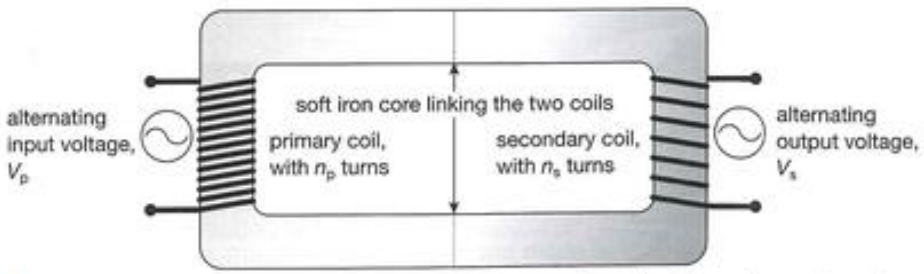
▲ Figure 21.13 This generator produces electricity on an industrial scale.

The size of the induced voltage is greater than in the bicycle dynamo because these generators **a** use much stronger magnets, **b** have many more turns of wire on the coil, and **c** spin the coil much faster.

PHYSICS ONLY

THE TRANSFORMER

When there is alternating current in a coil, the magnetic field around it is continuously changing. As the size of the current in the coil increases the field grows. As the size of the current decreases the field collapses. If a second coil is placed near the first, this growing and collapsing magnetic field will pass through it. As it cuts through the wires of the second coil, a voltage is induced across that coil. The size and direction of the induced voltage changes as the voltage applied to the first coil changes. (The first coil is more usually called the primary coil.) An alternating voltage applied across the primary coil therefore produces an alternating voltage across the secondary coil. This combination of two magnetically linked coils is called a transformer.



▲ Figure 21.14 The size of the voltage generated in the secondary coil of a transformer depends on the voltage in the primary coil and the numbers of turns on each coil.

REMINDER

Remember that transformers only work if the magnetic field around the primary coil is changing. Transformers will therefore only work with a.c. currents and voltages. They will not work with d.c. currents and voltages.

USING TRANSFORMERS TO CHANGE VOLTAGES

A **transformer** changes the size of an alternating voltage by having different numbers of turns on the input and output sides.

The relationship between the voltages across each of the coils is described by the equation:

$$\frac{\text{input voltage, } V_p}{\text{output voltage, } V_s} = \frac{\text{number of turns on primary coil, } n_p}{\text{number of turns on secondary coil, } n_s}$$

$$\frac{V_p}{V_s} = \frac{n_p}{n_s}$$

KEY POINT

A transformer that is used to increase voltages, like the one in Example 1, is called a step-up transformer. One that is used to decrease voltages is called a step-down transformer.

EXAMPLE 1

A transformer has 100 turns on its primary coil and 500 turns on its secondary coil. If an alternating voltage of 2 V is applied across the primary, what is the voltage across the secondary coil?

$$\frac{V_p}{V_s} = \frac{n_p}{n_s}$$

$$\frac{2\,\text{V}}{V_s} = \frac{100}{500}$$

$$V_s = \frac{500 \times 2\,\text{V}}{100}$$

$$= 10\,\text{V}$$

ENERGY IN TRANSFORMERS

It is very important to try to make transformers as efficient as possible. To keep any energy losses to a minimum we use thick copper wire for the coils and use a soft iron core covered in a thin layer of plastic to connect the two coils.

If a transformer is 100% efficient then the electrical energy entering the primary coil each second equals the electrical energy leaving the secondary coil each second: Another way of saying this is:

$$\text{input power, } V_pI_p = \text{output power, } V_sI_s$$

$$V_pI_p = V_sI_s$$

EXAMPLE 2

When a voltage of 12 V a.c. is applied across the primary coil of a step-down transformer, there is a current of 0.4 A in the primary coil. Calculate the current in the secondary coil if the voltage induced across it is 2 V a.c. Assume that the transformer is 100% efficient.

$$V_PI_P = V_SI_S$$

$$12\,\text{V} \times 0.4\,\text{A} = 2\,\text{V} \times I_S$$

$$I_S = \frac{12\,\text{V} \times 0.4\,\text{A}}{2\,\text{V}}$$

$$= 2.4\,\text{A}$$

TRANSFORMERS AND NATIONAL GRIDS

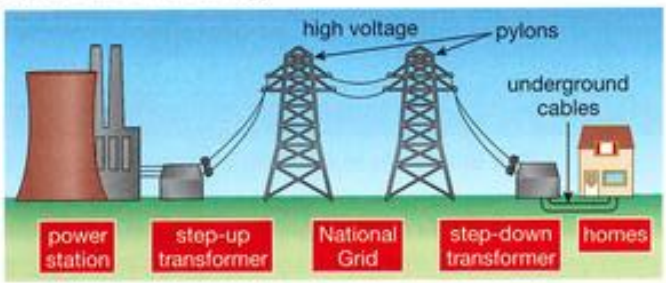


▲ Figure 21.15 A transformer

National Grids are networks of wires and cables that carry electrical energy from power stations to consumers such as factories and homes. Unfortunately, currents in long wires can lose lots of energy in the form of heat. The larger the current, the greater the amount of energy lost. If the current in the wires is kept to a minimum, the heat losses can be reduced. Transformers help us do this.

Transformers are used in National Grids so that the electricity is transmitted (sent out) as low currents and at high voltages.

Immediately after generation, electric currents from the alternators are passed through transformers which greatly decrease the size of the currents and increase the size of the voltages. These step-up transformers increase the voltage of the electricity to approximately 400 kV. High voltages like these can be extremely dangerous so the cables are supported high above the ground on pylons. As the cables enter towns and cities they are buried underground. Close to where the electrical energy is needed, the supply is passed through a step-down transformer that decreases the voltage to approximately 230 V, while at the same time increasing the current.

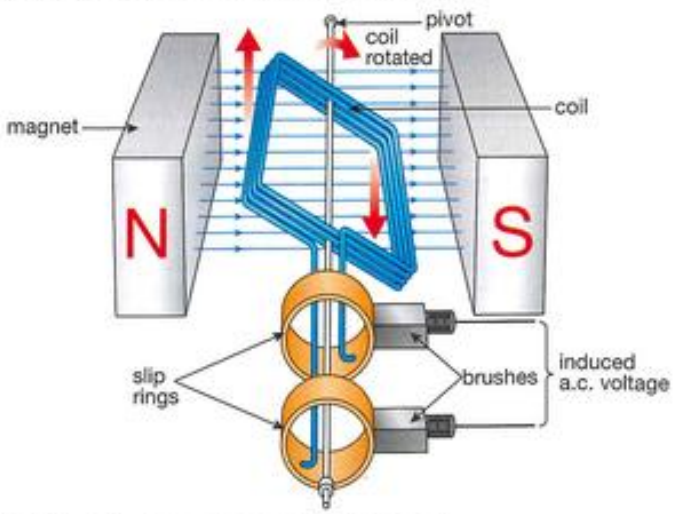


▲ Figure 21.16 Transformers are used in National Grids.

END OF PHYSICS ONLY

LOOKING AHEAD

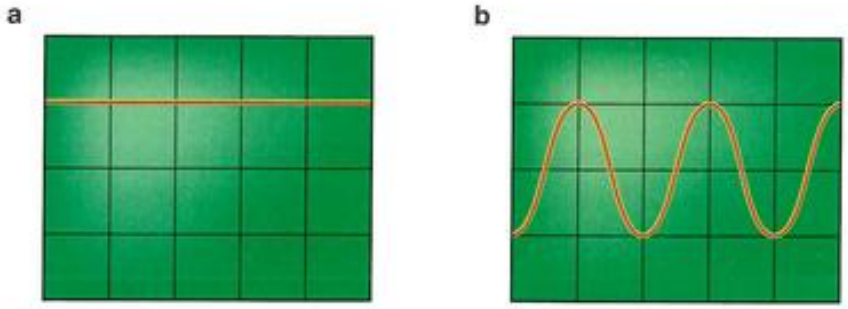
Higher Physics courses are likely to ask you what exactly is happening inside a generator and what is an alternator. The simple description given below will give you some idea of what is required.



▲ Figure 21.17 The alternator produces alternating current.

As the coil rotates, its wires cut through magnetic field lines and a current is induced in them. If we watch just one side of the coil we see that the wire moves up through the field and then down for each turn of the coil. As a result the current is induced first in one direction and then in the opposite direction. This kind of current is called alternating current. A generator that produces alternating current is called an alternator.

The frequency of an alternating current is the number of complete cycles it makes each second. If an alternator coil rotates twice in a second, the frequency of the alternating current it produces is 2 Hz (2 cycles per second). The frequency of the UK mains supply is 50 Hz.



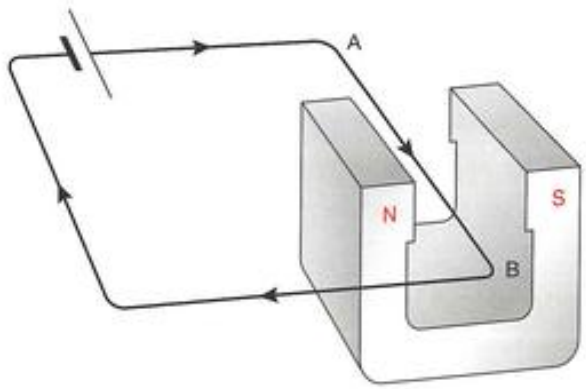
▲ Figure 21.18 a In a direct current (d.c) the charge flows in one direction. b Alternating currents (a.c.) keep changing direction.

CHAPTER QUESTIONS

More questions on electric motors and electromagnetic induction can be found at the end of Unit 6 on page 217.

SKILLS ANALYSIS, REASONING

1 The diagram below shows a long wire placed between the poles of a magnet.



Describe what happens when:

a there is current in the wire from A to B

b the direction of the current is reversed – that is, from B to A

c with the current from B to A, the poles of the magnet are reversed

d there is a larger current in the wire.

SKILLS INTERPRETATION

SKILLS CRITICAL THINKING

2 a Draw a simple labelled diagram of a moving-coil loudspeaker.

b Explain how sound waves are made by the speaker when a signal is passed through its coil.

PHYSICS ONLY

SKILLS CRITICAL THINKING (7)

3 a Explain the difference between a step-up transformer and a step-down transformer.

b Explain where step-up and step-down transformers are used in National Grids.

c Explain why transformers are used in National Grids.

SKILLS INTERPRETATION

d Draw a fully labelled diagram of a transformer.

SKILLS CRITICAL THINKING

e Explain why a transformer will not work if a d.c. voltage is applied across its primary coil.

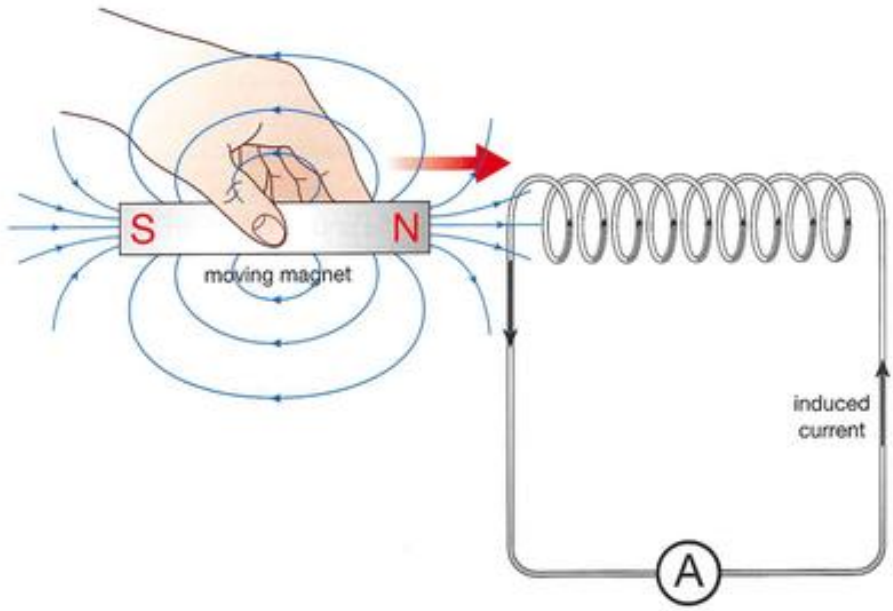
SKILLS PROBLEM SOLVING (8)

f A transformer has 200 turns on its primary coil and 5000 turns on its secondary coil. Calculate the voltage across the secondary coil when a voltage of 2 V a.c. is applied across the primary coil.

END OF PHYSICS ONLY

SKILLS REASONING (7)

4 The diagram below shows a bar magnet being pushed into a long coil. A sensitive meter is connected across the coil.



Describe what happens to the meter when:

a the magnet is pushed into the coil quickly

b the magnet is held stationary inside the coil

c the magnet is pulled out of the coil slowly

d the magnet is held stationary and the coil is moved towards it.